BDM Session 3

Correctness, convergence, and the evidence-gated path to validated physics

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The campaign was gated before it was scaled

3A — Correctness

Minimal safety patch only

Immutable legacy/corrected baselines

Synthetic, Fortran, identity, resource gates

Outcome: corrected numerical baseline green

3B — Finder + sampling

OFAT first; compact design frozen second

Same immutable state for finder variants

Nested and independent ID selections

Outcome: GO for bounded GR calibration

3C — End to end

Matched-phase particle and force scans

Finder mesh deconfounded from force mesh

Compact joint design; no Cartesian product

Outcome: bounded design closed; science NO-GO

Session 3C footprint: 22 new GR simulations | 23 physical states analysed | 135 terminal Slurm allocations | 49.12 billed core-hours

All ICs used validated FML 2LPTIC at $z_{\rm ini} = 49$ with stable uint64 IDs; all light work ran on `cosma8-serial`.

The corrected finder is numerically reproducible

Minimal Session 3 safety corrections

- Read and validate configuration before density or peak work
- Replace the raced fixed maxima buffer with stable compaction
- Emit a valid empty catalogue when there are zero maxima
- Use the declared snapshot background density
- Guard shape, $V_{\max}$, $R_{\max}$, roots, and denominators
- Record exact integer N_{bound} in an opt-in sidecar

Production physics, defaults, and legacy `Catshort` are unchanged.

21/21 final login-safe + real-Fortran tests pass

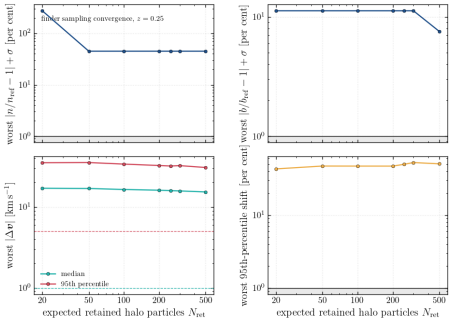
0 repeat/thread representation floor

1–128 OpenMP thread counts are byte-identical

fast = precise candidate sets and downstream propagation

Archived production executable unchanged:
`c84f4e3c…ad89`

Finder sampling is not converged by 500 particles



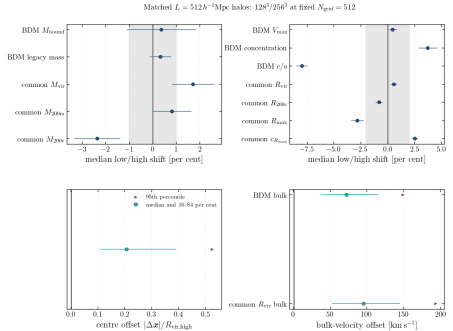
At the 500-particle cut

HMF shift + error	45.0%
clustering shift + error	7.54%
velocity median	$15.36\,\mathrm{km\,s^{-1}}$
velocity p95	$30.77\,\mathrm{km\,s^{-1}}$
internal-property p95	50.4%

All application thresholds remain unresolved above 500

Stable-ID construction passes. Label: **finder sampling convergence**, not simulation-resolution convergence.

Particle, force, and finder-mesh resolution are all sensitive



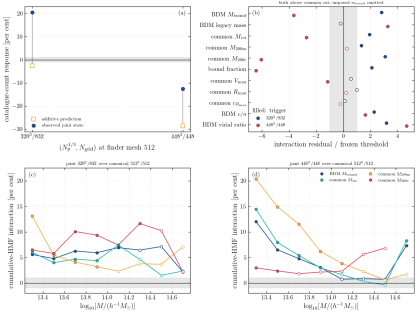
L512 cluster-tail control

$128^3/256^3$ particles at fixed $N_{\rm grid} = 512$; 786 reciprocal matches at the common mass cut.

common M_{200c}	-2.37%
BDM concentration	$+3.72\%$
BDM c/a	-7.95%
centre offset	$0.2075\,R_{\rm vir}$
BDM bulk offset	$72.2\,{\rm km\,s^{-1}}$
common-$R_{\rm vir}$ bulk offset	$95.5\,{\rm km\,s^{-1}}$

Particle resolution: sensitive (three phases). Pure force resolution: sensitive (one phase; finder mesh 512). Finder mesh: independently sensitive.

A separable particle–force response model fails



22,560	reciprocal matched halos
49/100	physical-property triggers
51/126	eligible HMF triggers

Catalogue-count response

$320^3/832$: observed $+20.52\%$, additive prediction -2.66%

$448^3/448$: observed -12.56%, additive prediction -28.68%

Largest cumulative-HMF residuals: 13.1%**,** 20.3%

Decision: finder GO; science resolution NO-GO

GO

Corrected canonical BDM is a numerically viable **GR finder**. Fast and precise builds remain viable; the bounded calibration design is complete.

NO-GO

Do not select a production particle/force resolution or launch the main science campaign. The f(R) campaign remains deferred.

Evidence limits

Thresholds remain > 500 particles. Force and compact designs have one phase; only $z = 0.25$ was tested.

Unavailable controls

No trusted external finder and no finder-specific HOD/HSM systematic allocation were found.

Unresolved physics

Modified-gravity bound mass has no accepted definition; f(R) therefore remains deferred.

Next proposed simulations: 0 | **No analytic-HMF tuning** | **Separate error budgets**

Session 4: physics changes stay opt-in until validated

legacy_original

Archived historical executable for exact reproduction only.

legacy_corrected

Session 3 safety baseline; unchanged legacy physics and schema.

validated_v1

Explicit opt-in mode for accepted Session 4 definitions and fields.

Evidence-backed implementation order

1. deterministic plateaus and stable peak keys;
2. one profile for $M_{\rm vir}$, $M_{\rm 200m}$, and $M_{\rm 200c}$;
3. centre, velocity, and iterative-unbinding comparisons;
4. periodic host flags plus resolved $V_{\rm max}$ and shapes;
5. versioned catalog schema and reader migration.

Acceptance loop for every patch

- one hypothesis and one reviewable behavior change;
- synthetic sentinels plus real-Fortran gates;
- immutable L100 cosmological comparison;
- reciprocal matching, HMF, exclusion, and velocity ablation;
- legacy identity; reject weaker invariance or definitions.

Adopt validated physics only when its full ablation is green.